

Determinants

Special Properties of Determinants

Let A be a square matrix. If the rows or columns of A are linearly dependent, then $\det(A) = 0$.

In particular, if two rows/columns of A are multiples of each other, then $\det(A) = 0$.

We also recover the fact that a matrix with a row or column of zeros has determinant zero.

EXAMPLE

The following matrices all have zero determinant:

$$\begin{pmatrix} 0 & 2 & -1 \\ 0 & 5 & 10 \\ 0 & -7 & 3 \end{pmatrix}, \quad \begin{pmatrix} 5 & -15 & 11 \\ 3 & -9 & 2 \\ 2 & -6 & 16 \end{pmatrix}, \quad \begin{pmatrix} 3 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 \\ 4 & 2 & 5 & 12 \\ -1 & 3 & 4 & 8 \end{pmatrix}, \quad \begin{pmatrix} \pi & e & 11 \\ 3\pi & 3e & 33 \\ 12 & -7 & 2 \end{pmatrix}.$$

Special Properties of Determinants

If A is a square matrix, then

$$\det(A^n) = \det(A)^n$$

EXAMPLE

Compute $\det(A^{100})$, where $A = \begin{pmatrix} 4 & 1 \\ 2 & 1 \end{pmatrix}$.

Solution

We have $\det(A) = 4 - 2 = 2$, so

$$\det(A^{100}) = \det(A)^{100} = 2^{100}.$$

Special Properties of Determinants

- Invertibility Property:

A square matrix is invertible if and only if $\det(A) \neq 0$.

- Multiplicativity Property:

If A and B are $n \times n$ matrices, then $\det(AB) = \det(A) \det(B)$.

- Transpose Property:

For any square matrix A , we have $\det(A^T) = \det(A)$.

Special Properties of Determinants

EXAMPLE

$$A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}.$$

SOLUTION

$$AB = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}$$

and

$$\det AB = 25 \cdot 13 - 20 \cdot 14 = 325 - 280 = 45$$

Since $\det A = 9$ and $\det B = 5$,

$$(\det A)(\det B) = 9 \cdot 5 = 45 = \det AB$$

Special Properties of Determinants

Transpose Property

The transpose property is very useful. For concreteness, we note that $\det(A) = \det(A^T)$ means, for instance, that

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \det \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}.$$

This implies that the determinant has the curious feature that it also behaves well with respect to *column* operations. Indeed, a column operation on A is the same as a row operation on A^T , and $\det(A) = \det(A^T)$.

Column Operations

Properties of a Determinant for Column Operations

The determinant satisfies the following properties with respect to the column operations:

- 1. Doing a column replacement on A does not change $\det(A)$.*
- 2. Scaling a column of A by a scalar c multiplies the determinant by c .*
- 3. Swapping two columns of a matrix multiplies the determinant by -1 .*

Column Operations

EXAMPLE Compute $\det \begin{pmatrix} 2 & 7 & 4 \\ 3 & 1 & 3 \\ 4 & 0 & 1 \end{pmatrix}$.

SOLUTION

It takes fewer column operations than row operations to make this matrix upper-triangular:

$$\begin{pmatrix} 2 & 7 & 4 \\ 3 & 1 & 3 \\ 4 & 0 & 1 \end{pmatrix} \xrightarrow{C_1 = C_1 - 4C_3} \begin{pmatrix} -14 & 7 & 4 \\ -9 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} \\ \xrightarrow{C_1 = C_1 + 9C_2} \begin{pmatrix} 49 & 7 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

Column Operations

SOLUTION (Continued...)

We performed two column replacements, which does not change the determinant; therefore,

$$\det \begin{pmatrix} 2 & 7 & 4 \\ 3 & 1 & 3 \\ 4 & 0 & 1 \end{pmatrix} = \det \begin{pmatrix} 49 & 7 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} = 49.$$

Column Operations

We can perform operations on the columns of a matrix in a way that is analogous to the row operations we have considered. The next theorem shows that column operations have the same effects on determinants as row operations.

If A is an $n \times n$ matrix, then $\det A^T = \det A$.

$$\begin{array}{ccc} & A & \\ \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} & \longrightarrow & \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix} \\ & & A^T \end{array}$$



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Cramer's Rule

Cramer's rule is needed in a variety of theoretical calculations. For instance, it can be used to study how the solution of $A\mathbf{x} = \mathbf{b}$ is affected by changes in the entries of $\mathbf{b}$.

For any $n \times n$ matrix A and any $\mathbf{b}$ in $\mathbb{R}^n$, let $A_i(\mathbf{b})$ be the matrix obtained from A by replacing column i by the vector $\mathbf{b}$.

$$A_i(\mathbf{b}) = [\mathbf{a}_1 \quad \cdots \quad \mathbf{b} \quad \cdots \quad \mathbf{a}_n]$$


col i

Let A be an invertible $n \times n$ matrix. For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \dots, n$$

Cramer's Rule

EXAMPLE Use Cramer's rule to solve the system

$$\begin{aligned}3x_1 - 2x_2 &= 6 \\ -5x_1 + 4x_2 &= 8\end{aligned}$$

SOLUTION View the system as $A\mathbf{x} = \mathbf{b}$. Using the notation introduced above,

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 4 \end{bmatrix}, \quad A_1(\mathbf{b}) = \begin{bmatrix} 6 & -2 \\ 8 & 4 \end{bmatrix}, \quad A_2(\mathbf{b}) = \begin{bmatrix} 3 & 6 \\ -5 & 8 \end{bmatrix}$$

Since $\det A = 2$, the system has a unique solution. By Cramer's rule,

$$\begin{aligned}x_1 &= \frac{\det A_1(\mathbf{b})}{\det A} = \frac{24 + 16}{2} = 20 \\ x_2 &= \frac{\det A_2(\mathbf{b})}{\det A} = \frac{24 + 30}{2} = 27\end{aligned}$$

Cramer's Rule

EXAMPLE Consider the following system in which s is an unspecified parameter. Determine the values of s for which the system has a unique solution, and use Cramer's rule to describe the solution.

$$\begin{aligned}3sx_1 - 2x_2 &= 4 \\ -6x_1 + sx_2 &= 1\end{aligned}$$

SOLUTION View the system as $A\mathbf{x} = \mathbf{b}$. Then

$$A = \begin{bmatrix} 3s & -2 \\ -6 & s \end{bmatrix}, \quad A_1(\mathbf{b}) = \begin{bmatrix} 4 & -2 \\ 1 & s \end{bmatrix}, \quad A_2(\mathbf{b}) = \begin{bmatrix} 3s & 4 \\ -6 & 1 \end{bmatrix}$$

Since

$$\det A = 3s^2 - 12 = 3(s + 2)(s - 2)$$

the system has a unique solution precisely when $s \neq \pm 2$.

Cramer's Rule

SOLUTION (Continued...)

For such an s , the solution is (x_1, x_2) , where

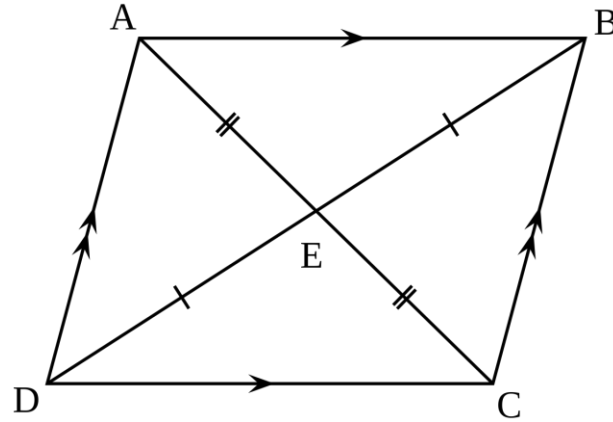
$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{4s + 2}{3(s + 2)(s - 2)}$$

$$x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = \frac{3s + 24}{3(s + 2)(s - 2)} = \frac{s + 8}{(s + 2)(s - 2)}$$

Determinants as Area and Volume

Parallelogram and Parallelopiped

Parallelogram is a two-dimensional geometrical shape whose opposite sides are equal in length and parallel. The opposite angles of a parallelogram are equal in measure.

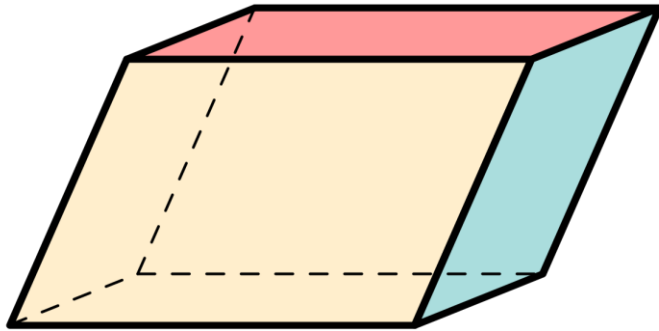


Note: All rectangles and squares can be considered parallelograms, but the opposite is not true, i.e. all parallelograms are not considered squares or rectangles.

Determinants as Area and Volume

Parallelogram and Parallelepiped (Continued...)

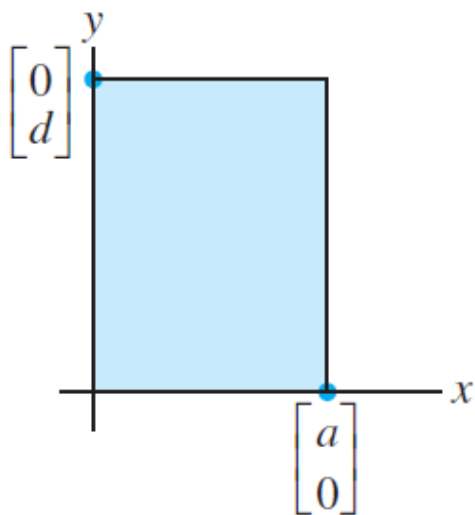
Parallelepiped is a three-dimensional structure whose every face is a parallelogram. A parallelepiped has 6 parallelogram faces.



Determinants as Area and Volume

Theorem

If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$.

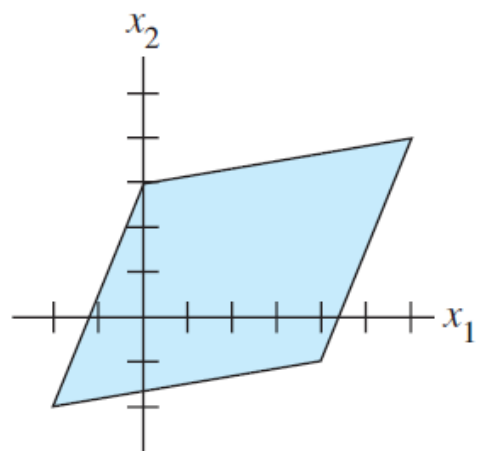


$$\left| \det \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \right| = |ad| = \left\{ \begin{array}{l} \text{area of} \\ \text{rectangle} \end{array} \right\}$$

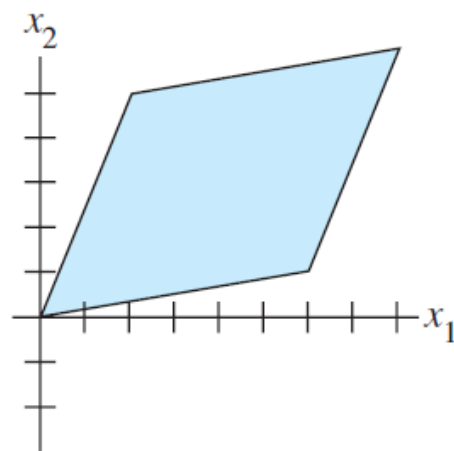
Determinants as Area and Volume

EXAMPLE Calculate the area of the parallelogram determined by the points $(-2, -2)$, $(0, 3)$, $(4, -1)$, and $(6, 4)$.

SOLUTION First translate the parallelogram to one having the origin as a vertex. For example, subtract the vertex $(-2, -2)$ from each of the four vertices. The new parallelogram has the same area, and its vertices are $(0, 0)$, $(2, 5)$, $(6, 1)$, and $(8, 6)$.



(a)



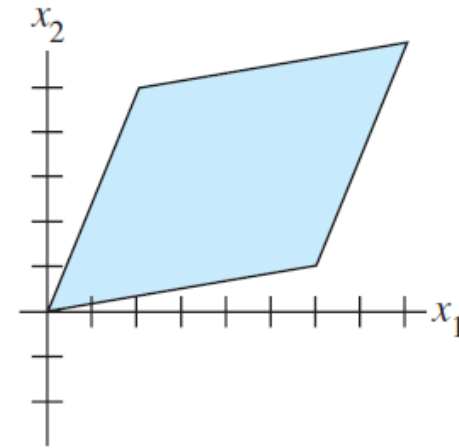
(b)

Determinants as Area and Volume

SOLUTION (Continued...)

This parallelogram is determined by the columns of

$$A = \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}$$

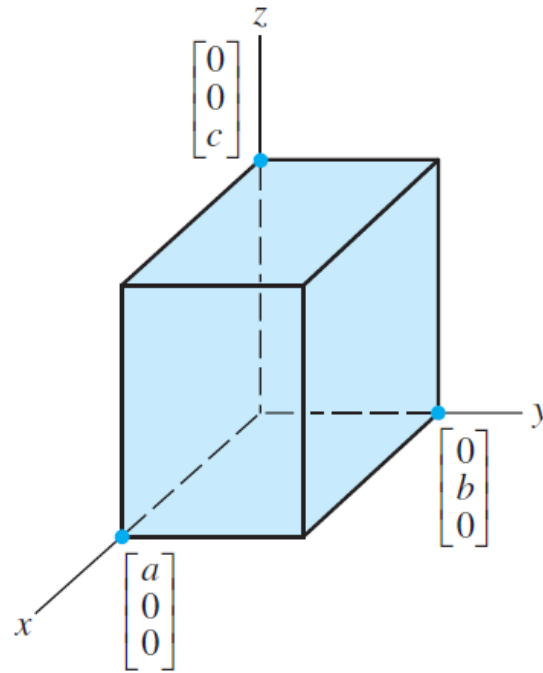


Since $|\det A| = |-28|$, the area of the parallelogram is 28.

Determinants as Area and Volume

Theorem

If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.



$$\text{Volume} = |abc|.$$

Determinants as Area and Volume

Properties of Determinants – Graphical Proof

Recall the four defining properties of determinants

1. Doing a row replacement on A does not change $|\det(A)|$.
2. Scaling a row of A by a scalar c multiplies $|\det(A)|$ by $|c|$.
3. Swapping two rows of a matrix does not change $|\det(A)|$.
4. The determinant of the identity matrix I_n is equal to 1.

$$|\det(A)| = \left| \frac{(\text{product of the diagonal entries of } B)}{(\text{product of scaling factors used})} \right|.$$

Determinants as Area and Volume

EXAMPLE Let a and b be positive numbers. Find the area of the region E bounded by the ellipse whose equation is

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$$

SOLUTION We claim that E is the image of the unit disk D under the linear transformation T determined by the matrix $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, because if $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, and $\mathbf{x} = A\mathbf{u}$, then

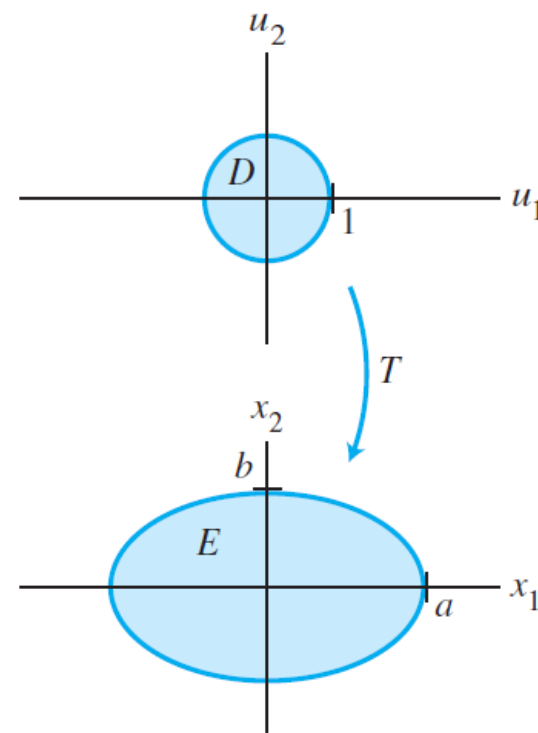
$$u_1 = \frac{x_1}{a} \quad \text{and} \quad u_2 = \frac{x_2}{b}$$

Determinants as Area and Volume

SOLUTION (Continued...)

It follows that $\mathbf{u}$ is in the unit disk, with $u_1^2 + u_2^2 \leq 1$, if and only if $\mathbf{x}$ is in E , with $(x_1/a)^2 + (x_2/b)^2 \leq 1$.

$$\begin{aligned}\{\text{area of ellipse}\} &= \{\text{area of } T(D)\} \\ &= |\det A| \cdot \{\text{area of } D\} \\ &= ab \cdot \pi(1)^2 = \pi ab\end{aligned}$$



Determinants as Area and Volume

EXAMPLE

Let S be the parallelogram determined by the vectors $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$. Compute the area of the image of S under the mapping $\mathbf{x} \mapsto A\mathbf{x}$.

SOLUTION

The area of S is $\left| \det \begin{bmatrix} 1 & 5 \\ 3 & 1 \end{bmatrix} \right| = 14$, and $\det A = 2$.

The area of image of S under the mapping $\mathbf{x} \mapsto A\mathbf{x}$ is

$$|\det A| \cdot \{\text{area of } S\} = 2 \cdot 14 = 28$$